Auditable Forgetting: Membership-Inference Certificates for Verifying Machine Unlearning under Data-Protection Mandates

Shardul Chogale
Vidyalankar Institute of Technology
Mumbai, India
shardul.chogale@vit.edu.in

***Abstract*—Under India's Digital Personal Data Protection (DPDP) Act and the EU's GDPR, a person can ask a company to delete their data. If a model was trained on that data, deleting the rows does not help much; the model still carries traces of it. Removing those traces is called machine unlearning. Full retraining works but is slow and costly, so companies tend to use faster approximations, and an outsider has no easy way to tell whether these actually forget anything. This paper describes an audit that needs only the model's outputs. The auditor scores the deleted records and a set of records the model has never seen, then asks whether the two groups can be told apart. We turn the answer into a lower bound $\hat{\varepsilon}_{\mathrm{LB}}$ using Clopper–Pearson intervals, and add a Kolmogorov–Smirnov comparison against a retrained model. We tried this on synthetic data with logistic regression. Fine-tuning and gradient ascent did almost nothing: $\hat{\varepsilon}_{\mathrm{LB}}$ stayed near 0.42, its value before deletion, and test accuracy did not move from 67.6%. A Newton step with added noise brought $\hat{\varepsilon}_{\mathrm{LB}}$ down to about 0.18, though the result changed a lot between seeds. Since no weights are needed, a regulator could run this check without the company's cooperation beyond query access.**

***Index Terms*—machine unlearning, membership inference, privacy auditing, differential privacy, right to erasure, responsible AI**

I. Introduction

Most deployed models are trained on data that users supplied. Once a model is live, a user may ask for their records to be erased under Section 12 of India's DPDP Act [1] or Article 17 of the GDPR [2]. Removing the rows from a database leaves the trained weights untouched, and those weights can still give away which records were used: membership inference attacks recover this information from model outputs alone [3], [4]. Machine unlearning [5]–[7] tries to edit the weights so the model behaves as if it had been trained on the remaining data only.

Retraining after every request is the exact answer, but it costs a full training run each time. In practice, operators use cheaper updates instead: a few more gradient steps on the retained data, gradient ascent on the deleted data, or a single Newton step computed from the Hessian [8]–[10]. These shortcuts can leave much of the deleted data's influence in place, and ordinary accuracy checks will not catch it [11], [12]. Operators and outside auditors therefore need a way to test unlearning from the model's outputs.

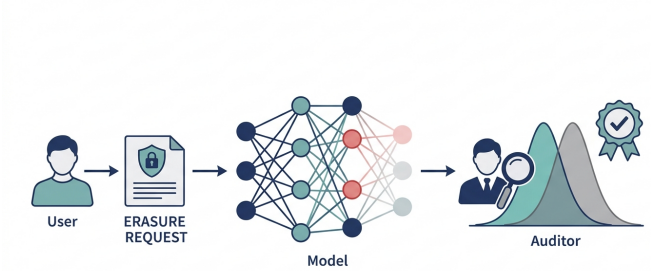


Fig. 1. Overall workflow. A user asks for deletion, the operator runs an unlearning method, and an independent auditor compares the model's scores on the deleted records with its scores on records it never saw.

Contributions. Fig. 1 shows the setting. Our contributions are:

- a definition of (ε, δ)-membership-indistinguishable forgetting, stated in terms of the true- and false-positive rates of a membership attack (Section III);
- a lower bound $\hat{\varepsilon}_{\mathrm{LB}}$ that holds with confidence $1 - \alpha$, built from exact Clopper–Pearson intervals on per-sample losses, plus a distribution test against a retrained model when one is available (Section IV);
- open-source experiments comparing fine-tuning, gradient ascent and a noisy Newton update over several deletion sizes and noise levels (Sections V–VI);
- a discussion of what an auditor needs in practice and where the test falls short (Section VII).

II. Related Work

Machine unlearning. Cao and Yang [5] rewrote learning algorithms in a summation form so that removing a point means updating a few sums. SISA [6] trains separate models on disjoint shards, so a deletion only forces one shard to be retrained. Other work covers k-means clustering [13], convex models with provable removal guarantees [8], [10], and deep networks, where Golatkar et al. scrub weights using the Fisher information [9]. Most formal definitions are modeled on differential privacy (DP) [14].

Membership inference. Shokri et al. [3] trained shadow models to tell training points from other points using a model's confidence scores. Yeom et al. [4] showed that simply thresholding the loss works too, and tied the attack's success to overfitting. Carlini et al. [15] argued that attacks should be judged at low false-positive rates, since that is where memorization of individual records shows up.

Privacy auditing. Jagielski et al. [16] and Nasr et al. [17] checked DP training code by running attacks and turning their success rates into lower bounds on ε. Steinke et al. [18] showed how to do this with a single training run. We borrow the same hypothesis-testing idea, but instead of planting canaries in neighbouring datasets, we compare the deleted records against fresh records from the same distribution.

III. Problem Formulation and Threat Model

Let $\mathcal{D} = \{z_i = (x_i, y_i)\}_{i=1}^{n}$ be a training set drawn i.i.d. from a distribution P, and let $\theta = A(\mathcal{D})$ be the trained weights. A deletion request names a subset $\mathcal{D}_f \subset \mathcal{D}$; the rest is $\mathcal{D}_r = \mathcal{D} \setminus \mathcal{D}_f$. An unlearning method $\mathcal{U}$ maps $(\theta, \mathcal{D}_f, \mathcal{D}_r)$ to new weights θ_u. Retraining gives the reference weights $\theta_r = A(\mathcal{D}_r)$.

What the auditor has. The auditor (a regulator or a data protection officer, say) holds the forget set $\mathcal{D}_f$ and a separate pool $\mathcal{D}_{\mathrm{n}} \sim P^m$ with $\mathcal{D}_{\mathrm{n}} \cap \mathcal{D} = \emptyset$. They can query θ_u and read one number per record. We use the negative loss:

$$s_\theta(z) = -\ell(\theta; z), \qquad \ell(\theta; (x, y)) = \log\Big(1 + e^{\theta^\top x}\Big) - y\,\theta^\top x, \tag{1}$$

as in the loss-threshold attack of [4]. Any other score computed from the outputs would also work.

Idea. The retrained model θ_r never saw $\mathcal{D}_f$, so to θ_r the deleted records look just like the fresh records in $\mathcal{D}_{\mathrm{n}}$: their scores come from the same distribution. If θ_u still scores $\mathcal{D}_f$ differently from $\mathcal{D}_{\mathrm{n}}$, the deleted data has not been fully removed.

Definition 1 (Membership-indistinguishable forgetting)**.** *An algorithm $\mathcal{U}$ achieves (ε, δ)-membership-indistinguishable forgetting for score function s if, for any decision threshold τ and both tail directions,*

$$\Pr_{z \sim \mathcal{D}_f}[s_{\theta_u}(z) \geq \tau] \leq e^{\varepsilon} \Pr_{z \sim P}[s_{\theta_u}(z) \geq \tau] + \delta. \tag{2}$$

Kairouz et al. [19] give DP the same reading in terms of hypothesis tests. For an attacker, it says that at every threshold τ the true-positive rate is at most $e^{\varepsilon}\,\mathrm{FPR}(\tau) + \delta$. A model retrained without $\mathcal{D}_f$ meets it with $\varepsilon = \delta = 0$.

IV. Auditing Protocol and Certificate Construction

A. Confidence-Bounded Verification

Let $n_1 = |\mathcal{D}_f|$ and $n_0 = |\mathcal{D}_{\mathrm{n}}|$, and for a threshold τ let $k_1(\tau)$ and $k_0(\tau)$ be the number of records in $\mathcal{D}_f$ and $\mathcal{D}_{\mathrm{n}}$ with $s(z) \geq \tau$. Plugging raw rates into the definition would let sampling noise inflate the estimate, so we take the conservative end of exact Clopper–Pearson intervals [20]:

$$\underline{\mathrm{TPR}}(\tau) = B^{-1}\big(\tfrac{\alpha}{2};\, k_1,\, n_1 - k_1 + 1\big), \tag{3}$$

$$\overline{\mathrm{FPR}}(\tau) = B^{-1}\big(1 - \tfrac{\alpha}{2};\, k_0 + 1,\, n_0 - k_0\big), \tag{4}$$

where $B^{-1}(\cdot; a, b)$ is the beta quantile function. The bound is then

$$\hat{\varepsilon}_{\mathrm{LB}} = \max_{\tau \in \mathcal{T}} \max\left\{0,\ \ln \frac{\underline{\mathrm{TPR}}(\tau) - \delta}{\overline{\mathrm{FPR}}(\tau)}\right\}, \tag{5}$$

with τ ranging over a grid of quantiles $\mathcal{T}$ and both tails checked.

Proposition 1. *Suppose $\mathcal{U}$ provides (ε, δ)-membership-indistinguishable forgetting. For a fixed decision threshold τ, the lower bound satisfies $\Pr[\hat{\varepsilon}_{\mathrm{LB}}(\tau) > \varepsilon] \leq \alpha$.*

Proof sketch. Each of (3) and (4) is a one-sided bound at level $1 - \alpha/2$. By a union bound, $\underline{\mathrm{TPR}} \leq \mathrm{TPR}$ and $\overline{\mathrm{FPR}} \geq \mathrm{FPR}$ both hold with probability at least $1 - \alpha$. When they do, (2) gives

$$\underline{\mathrm{TPR}} - \delta \leq \mathrm{TPR} - \delta \leq e^{\varepsilon}\mathrm{FPR} \leq e^{\varepsilon}\overline{\mathrm{FPR}},$$

so $\ln[(\underline{\mathrm{TPR}} - \delta)/\overline{\mathrm{FPR}}] \leq \varepsilon$. □

Remark 1. *Taking the maximum over $|\mathcal{T}|$ thresholds needs a Bonferroni correction, $\alpha \to \alpha/(2|\mathcal{T}|)$, for the guarantee to hold. Our experiments report the uncorrected value, since we use it to compare methods rather than to certify any one of them.*

B. Reference Divergence Metrics

If the auditor can also train θ_r (for instance in a sandbox), we compare the two models directly with the two-sample Kolmogorov–Smirnov statistic

$$\mathrm{KS} = \sup_t \left| F_{\theta_u}^{\mathcal{D}_f}(t) - F_{\theta_r}^{\mathcal{D}_f}(t) \right|, \tag{6}$$

where $F_\theta^{\mathcal{D}_f}$ is the empirical CDF of the scores on $\mathcal{D}_f$. We also report the weight distance $\Delta_\theta = \|\theta_u - \theta_r\|_2$.

C. Newton Removal with Additive Noise

For the regularized objective $J(\theta; \mathcal{D}) = \sum_{z \in \mathcal{D}} \ell(\theta; z) + \frac{\lambda n}{2}\|\theta\|^2$, one Newton step from the optimum $\theta^\star$ estimates the effect of dropping $\mathcal{D}_f$:

$$\theta_u = \theta^\star + H_{\mathcal{D}_r}^{-1}\left(\sum_{z \in \mathcal{D}_f} \nabla \ell(\theta^\star; z) - \lambda |\mathcal{D}_f| \theta^\star\right) + \boldsymbol{\xi}, \tag{7}$$

where $H_{\mathcal{D}_r} = \nabla^2 J(\theta^\star; \mathcal{D}_r)$ and $\boldsymbol{\xi} \sim \mathcal{N}(0, \sigma^2 I)$, following Guo et al. [8]. The noise is there to hide the error left over from the second-order approximation.

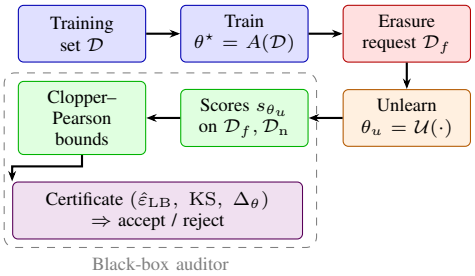


Fig. 2. The audit pipeline. The auditor queries the loss on each deleted record and each fresh record, then turns the counts into confidence bounds.

Algorithm 1 Forgetting Certificate Evaluation

Require: Unlearned model θ_u, forget set $\mathcal{D}_f$, holdout instances $\mathcal{D}_n$, threshold set $\mathcal{T}$, confidence level α, relaxation δ

Extract loss scores: $S_1 \leftarrow \{s_{\theta_u}(z) : z \in \mathcal{D}_f\}$, $S_0 \leftarrow \{s_{\theta_u}(z) : z \in \mathcal{D}_n\}$
Initialize metric: $\hat{\varepsilon}_{LB} \leftarrow 0$
for $\tau \in \mathcal{T}$ and search direction $d \in \{+1, -1\}$ **do**
    Count tail events: $k_1 \leftarrow |\{s \in S_1 : d\,s \geq d\,\tau\}|$, $k_0 \leftarrow |\{s \in S_0 : d\,s \geq d\,\tau\}|$
    Evaluate bounds $\underline{\mathrm{TPR}}$ and $\overline{\mathrm{FPR}}$ using (3)–(4)
    Update bound: $\hat{\varepsilon}_{LB} \leftarrow \max\{\hat{\varepsilon}_{LB}, \ln[(\underline{\mathrm{TPR}} - \delta)/\overline{\mathrm{FPR}}]\}$
end for
return $\hat{\varepsilon}_{LB}$ ▷ Flag unlearning violation if $\hat{\varepsilon}_{LB} > \varepsilon_{policy}$

V. Experimental Configuration

Data. All data here is synthetic, which keeps real personal records out of the study and lets others repeat our runs. Each example has $d = 300$ Gaussian features and a binary label drawn from a random logistic model with label noise. There are $n = 600$ training examples, 3000 held-out examples for the auditor and 2000 test examples. With d this large relative to n, the model overfits and memorizes individual points, which is what we want to detect.

Model. L_2-regularized logistic regression ($\lambda = 10^{-4}$), trained with Newton–Raphson until the gradient vanishes.

Methods. We compare five models: (i) *Original*, the model before deletion; (ii) *Retrain*, trained from scratch on $\mathcal{D}_r$; (iii) *Fine-tune*, five gradient descent steps on $\mathcal{D}_r$; (iv) *Grad-ascent*, five ascent steps on $\mathcal{D}_f$ then two descent steps on $\mathcal{D}_r$; and (v) *Newton*, the update in Eq. (7) with $\sigma = 0.02$.

Settings. Unless stated otherwise, $|\mathcal{D}_f|/n = 10\%$ (60 deleted records), $|\mathcal{D}_n| = 3000$, $\alpha = 0.05$, $\delta = 0$ and $|\mathcal{T}| = 99$. We report mean and standard deviation over 10 seeds. The code is in `scripts/simulate.py`.

VI. Empirical Findings

Main comparison. Table I and Fig. 3 give the results. Before deletion, the attack can tell the forget set apart from fresh data (AUC $= 0.606$, $\hat{\varepsilon}_{LB} = 0.421$), so the model has memorized its training points. Retraining brings the AUC down to 0.504 and $\hat{\varepsilon}_{LB}$ to 0.005, as it should.

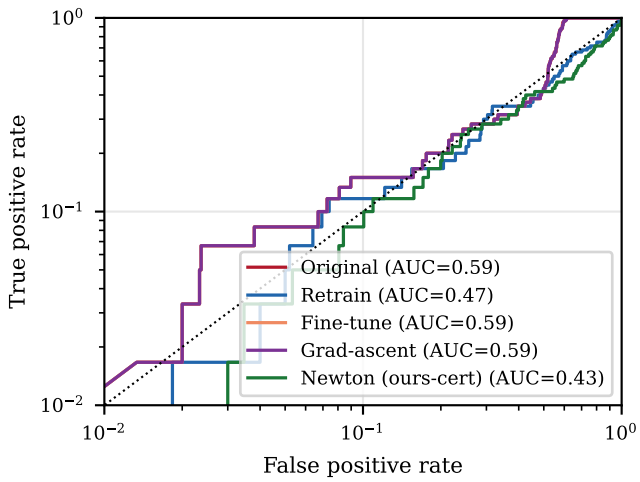


Fig. 3. ROC curves (log scale) for the loss-threshold attack separating deleted records from fresh ones, seed 0. A curve on the diagonal means the attack does no better than chance.

Fine-tuning and gradient ascent score almost exactly like the original model ($\hat{\varepsilon}_{LB} = 0.419$, AUC $= 0.606$, KS $= 0.398$), and test accuracy stays at 67.6%. With so few steps the weights barely move ($\Delta_\theta \approx 6.17$, against 6.18 for the original), so accuracy alone would make these methods look as if they had worked. The noisy Newton update lowers $\hat{\varepsilon}_{LB}$ to 0.179 and KS to 0.147, a 55% drop, at the cost of some accuracy (65.3%). Its spread across seeds is large (± 0.249) and it stays far from the retrained weights ($\Delta_\theta = 4.53$), so one Newton step is not a reliable stand-in for retraining.

Score distributions. Fig. 4 shows why accuracy misses the problem. The fine-tuned model is as accurate as before, yet its losses on the deleted records are still visibly lower than on fresh records. The KS column in Table I measures this gap against the retrained model.

Noise level. Fig. 5 varies σ. Up to $\sigma = 0.02$, $\hat{\varepsilon}_{LB}$ sits near 0.245. It falls to its lowest value, 0.126, at $\sigma = 0.1$, then rises again at $\sigma = 0.2$, where test accuracy has dropped to about 65.0%. With only 60 deleted records, the error of the quadratic approximation matters more than the added noise. Because a single seed can mislead, auditors should look at results over several seeds.

Deletion size. Fig. 6 shows two trends. For the original model and the cheap updates, $\hat{\varepsilon}_{LB}$ rises from 0.25 at 2.5% deletion to 0.46 at 30%. The leakage is not growing; the intervals simply get tighter as the forget set grows. Retraining stays at $\hat{\varepsilon}_{LB} \leq 0.048$ throughout. The Newton update does well for small deletions ($\hat{\varepsilon}_{LB} = 0.09$ at 5%) but poorly for large ones (0.67 at 20%), because the weights move too far for a local Taylor approximation to hold. In short, single requests are cheap to process but need a large fresh pool to audit, while large batches are better handled by sharded retraining [6].

TABLE I

AUDIT PERFORMANCE ACROSS UNLEARNING BASELINES (MEAN ± STD ACROSS 10 TRIALS, $|\mathcal{D}_f|/n = 10\%$). LOWER $\hat{\varepsilon}_{\mathrm{LB}}$, KS AND Δ_θ, AND MIA AUC CLOSER TO 0.5, MEAN BETTER FORGETTING.

Method	Test acc. ↑	MIA AUC → 0.5	$\hat{\varepsilon}_{\mathrm{LB}}$ ↓	KS to retrain ↓	Δ_θ ↓
Original	0.676±0.020	0.606±0.016	0.421±0.025	0.398±0.061	6.179±0.694
Retrain	0.675±0.014	0.504±0.033	0.005±0.010	0.000±0.000	0.000±0.000
Fine-tune	0.676±0.020	0.606±0.016	0.419±0.027	0.398±0.061	6.175±0.693
Grad-ascent	0.676±0.020	0.606±0.016	0.419±0.027	0.398±0.061	6.174±0.693
Newton (ours-cert)	0.653±0.015	0.482±0.041	0.179±0.249	0.147±0.035	4.529±1.053

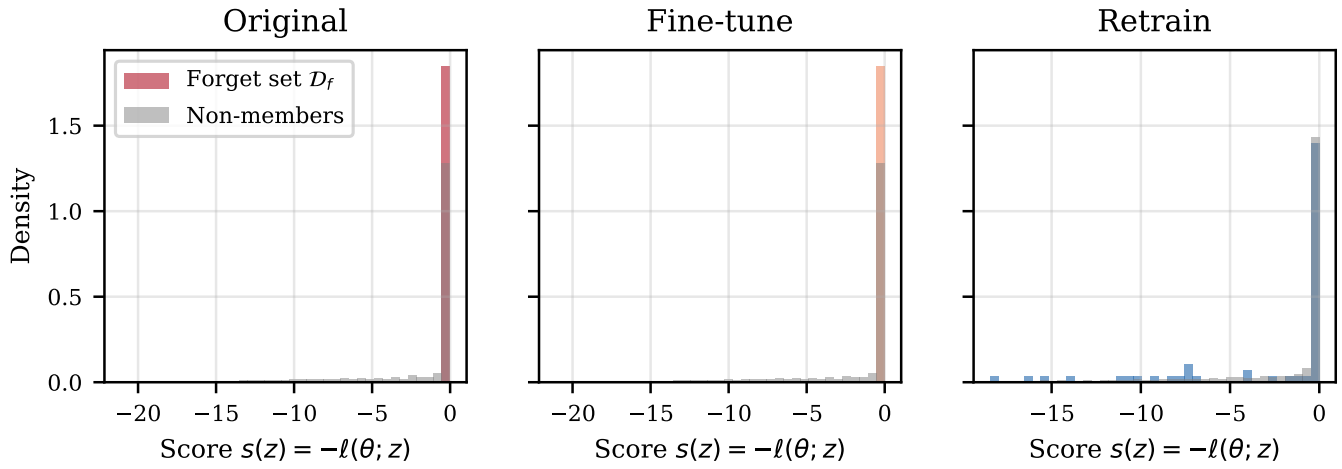


Fig. 4. Score distributions $s_\theta(z)$ for deleted records and fresh records. After fine-tuning the two still differ, even though test accuracy gives no sign of it.

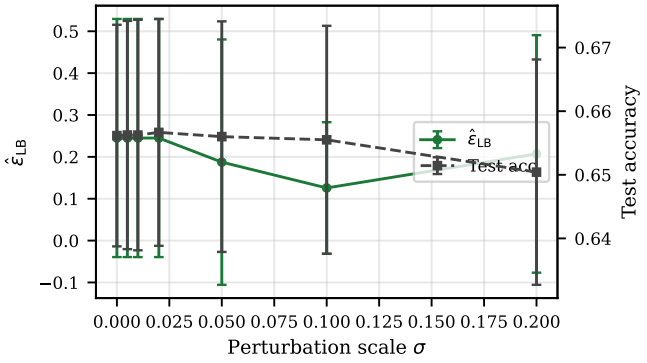


Fig. 5. Effect of the noise level σ in the Newton update on $\hat{\varepsilon}_{\mathrm{LB}}$ (left axis) and test accuracy (right axis), 6 seeds.

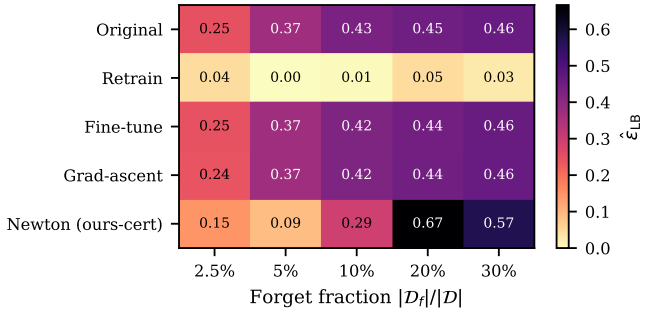


Fig. 6. Mean $\hat{\varepsilon}_{\mathrm{LB}}$ for each method and deletion fraction, 5 seeds. Larger forget sets give tighter intervals and so a more sensitive audit.

VII. AUDITOR CONSIDERATIONS AND ETHICAL CONTEXT

The bound only covers the attack we ran. A small $\hat{\varepsilon}_{\mathrm{LB}}$ says the loss-threshold attack fails; a stronger attack might still succeed. We also tested only a convex model on synthetic data, and deep networks and shadow-model attacks [15] need to be checked before this is used in production. The auditor must hold $\mathcal{D}_f$, which is exactly the data the user asked to delete, so the audit should run in a secure enclave and the copy should be destroyed afterwards. Last, a passing result is one piece of technical evidence. It does not replace the records and processes that data protection law already requires of an organization.

VIII. CONCLUSION

We turned a membership inference attack into a black-box test of machine unlearning with a confidence guarantee. In our experiments, a few steps of fine-tuning or gradient ascent left the deleted data as visible as before ($\hat{\varepsilon}_{\mathrm{LB}} \approx 0.42$) while accuracy looked normal. A noisy Newton step helped for small deletions but not for large ones. Next, we plan to test neural networks, try single-run auditing [18], and connect the test to how DPDP deletion requests are handled in practice.

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